Hima Shanthi Rayapati

hrayapa1@asu.edu | 623-444-0816 | LinkedIn | GitHub | Portfolio

Education

Arizona State University, MS in Psychology (Cognition, Behavior, and Information) 08/2026 – 12/2027

Arizona State University, BS in Psychology and Mathematics (Statistics) 01/2023 – 05/2026

- **Minor:** Philosophy **GPA:** 4.0/4.0

Research Experience

Research Assistant, *Thinking Across Languages and Contexts Lab*, ASU 08/2025 – Present

- Assist a graduate student with testing experiments built in jsPsych and JavaScript.
- Help check whether data are recorded correctly and organized for analysis.
- Perform exploratory data analysis and assist in interpreting experimental results under supervision.

Research Assistant, *Perception, Action, Language & Meaning Lab*, ASU 08/2025 – Present

- Run two-person collaborative task sessions (e.g., Taboo), observing and recording coordination, communication, and performance data using VR, motion-capture, and eye-tracking.
- Work with the lab's time-series behavioral data, learning how to clean, organize, and explore it.
- Read papers and explore possible methods (e.g., multilevel vector autoregression) to understand which approaches might fit the lab's data.
- Contribute to discussions by sharing preliminary notes and observations from trial analyses.

Volunteer Research Assistant, *Longitudinal & Behavioral Data Science Lab*, ASU 08/2024 – 12/2024

- Participated in bi-weekly lab meetings, contributing to discussions and staying updated on research progress.

Research Assistant, *Evolution, Ecology, and Social Behavior Lab*, ASU 08/2023 – 11/2023

- Assisted with survey design, pilot testing, data coding, and literature reviews in social cognition research.
- Participated in lab meetings and contributed to discussions on study design and interpretation.

Projects

Analyzing the Impact of AI on Stack Overflow Participation (NLP & Data Analysis) ***GitHub***

- Analyzed large-scale Stack Overflow data using Python to study changes in user behavior after AI tool releases.
- Built time-series datasets and computed engagement metrics (answer rates, response time, retention) using Python.
- Performed statistical comparisons and visualized platform-level trends.
- Explored tag-based trends and basic NLP features to examine topic shifts.

Stochastic Dual-Timescale Model of Learning (ODE & SDE Modeling) ***Final Project / 2025***

- Developed differential equation model of fast and slow learning processes.
- Extended to stochastic framework to examine effects of noise.
- Verified theoretical properties via simulation.
- Linked model behavior to memory retention and learning variability.

Cultural Differences in Trust: AI vs. Human Expert Decision-Making ***Research Proposal / 2025***

- Proposed cross-cultural study (Mexico vs. U.S.) on AI vs. human decision-making under conflict.
- Outlined experimental design focusing on competence and legitimacy.
- Formulated mixed-effects modeling approach for analysis.
- Synthesized literature across cognition, decision-making, and human-AI interaction.

Probabilistic Reward Learning Task using PsychoPy ***Reward Learning.com/2025***

- Designed probabilistic reward-learning task with informative vs. non-informative features.

- Examined learning under uncertainty and perceptual noise.
- Collected and analyzed participant data using Python.

Trust in Advice: AI vs. Human Sources (Data Analysis & Written Report) ***Trust in AI.com/2025***

- Analyzed 30,000+ human-AI interactions to study effects of advice source.
- Modeled decision changes before and after advice exposure.
- Found advice correctness as primary driver, with modest AI-related effects.

Research Proposal: Mirror Neurons, Social Synchrony, and Collaborative Learning ***Proposal.com/2025***

- Proposed a research framework synthesizing human and animal literature on mirror neuron systems and social synchrony.
- Formulated hypotheses linking coordination mechanisms to collaborative learning outcomes.

Additional projects and full write-ups available at my portfolio.

Work Experience

Research Aide, ASU – Tempe, AZ 12/2024 – Present

- Review, categorize, and structure user-generated data to ensure quality and consistency.
- Assign relevant tags to comments for improved organization and searchability.
- Evaluate AI-generated tags against human annotation standards, analyzing human-AI agreement and error patterns across prompt versions to guide model refinement.
- Perform data cleaning tasks, identifying and correcting inconsistencies to maintain data accuracy.
- Work with databases to structure data and ensure accessibility for future use.

Lead Survey Assistant, ASU – Tempe, AZ 05/2023 – Present

- Conduct phone-based survey interviews with currently employed and recently graduated ASU students, ensuring accurate and comprehensive data collection by clarifying responses and addressing any ambiguities.
- Assist in the hiring process, including interviewing and evaluating candidates.
- Perform data cleaning and ensure the integrity of datasets for analysis.
- Train new student workers in survey procedures and oversee testing and calibration of survey tools.

Intern, Public Safety Crisis Solution – Phoenix, AZ 08/2024 – 11/2024

- Performed quantitative analysis to support research, program development, and accreditation documentation.
- Contributed research-based content on clinical procedures addressing first responders' mental health initiatives.

Advising Office Aide, ASU – Tempe, AZ 08/2023 – 05/2024

- Supported advising operations through scheduling, communication, and administrative workflows.
- Managed data entry, marketing/outreach materials, and front-desk support for students and staff.

Skills

- **Programming & Statistical Tools:** R, SAS, MATLAB, Python, JMP, JASP, SQL (Microsoft Access), Java, C, Bash, REST APIs
- **Machine Learning & NLP (Foundational):** Principal Component Analysis (PCA), Neural network basics, Supervised vs. Unsupervised learning, Model evaluation concepts, Introductory text analysis (Voyant Tools)
- **Experimental & Research Tools:** jsPsych, PsychoPy, JavaScript, HTML, CSS, Qualtrics, SurveyMonkey
- **Data Analysis & Visualization:** Tableau, Excel, Google Sheets, Alteryx
- **Collaboration & Productivity:** Slack, Teams, Zoom, Word, Outlook, Adobe PowerPoint, OneDrive, Dropbox